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Elucidation of superionic conduction in K_2NiF_4 -type Ba–Li oxyhydride from first-principles molecular dynamics study

Jun Haruyama,^{*,†} Fumitaka Takeiri,^{†,‡} Genki Kobayashi,^{†,§} and Osamu Sugino[†]

[†]Solid State Chemistry Laboratory, Pioneering Research Institute (PRI), RIKEN, 2-1 Hirosawa, Wako, Saitama 351-0198, Japan

[‡]Department of Chemistry, Kindai University, 3-4-1 Kowakae, Higashiosaka, Osaka 577-8502, Japan

[§]Department of Electrical Engineering and Bioscience, School of Advanced Science and Engineering, Waseda University, 3-4-1 Okubo, Shinjuku-ku, Tokyo 169-8555, Japan

[†]The Institute for Solid State Physics, The University of Tokyo, 5-1-5 Kashiwanoha, Kashiwa, Chiba 277-8581, Japan

*Corresponding author

E-mail: jun.haruyama@riken.jp



ABSTRACTView Article Online
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Hydride-ion (H^-) conductors have attracted considerable attention as charge carriers for hydrogen transport applications. To understand the microscopic H^- dynamics and elucidate the superionic conduction characteristics, we investigated K_2NiF_4 -type Ba–Li oxyhydrides, $\text{Ba}_{1.75}\text{LiH}_{2.7}\text{O}_{0.9}$ (BLHO). Three microscopic structural models were constructed: $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$, $\text{Ba}_{57}\text{Li}_{32}\text{H}_{90}\text{O}_{28}$, and $\text{Ba}_{58}\text{Li}_{32}\text{H}_{90}\text{O}_{29}$, in which the H and O atoms at the apical sites were ordered in two ways (β - and δ -phase-like). By employing first-principles electronic structure calculations, the constructed models showed band gaps of 2.2–2.7 eV, occupied H-1s orbitals, and Bader charges of $\text{Ba}^{1.37+}$, $\text{Li}^{0.85+}$, $\text{H}^{0.73-}$, and $\text{O}^{1.45-}$, which is consistent with the pure H^- conduction and electrochemical H^- carriers observed in previous studies. Using first-principles molecular dynamics (FPMD) simulations, the H^- diffusion coefficients of the β - and δ -like models were in good agreement with those reported in previous quasi-elastic neutron scattering analyses. The activation energies of the *ab*-plane (0.52 and 0.55 eV in the β and δ models, respectively) and *c*-axis components (0.78 eV in the δ model) from FPMD simulations were found to be related to migration barriers of distinct H^- transfer pathways. Furthermore, the Lindemann index of H exceeded the threshold for a solid–liquid transition at temperatures above 700 K, suggesting sublattice melting of the H^- equatorial sites. Finally, above the sublattice melting temperature ($T \geq 700$ K), the van Hove correlation function showed the peak growth at correlation hole position increased for few picoseconds, and rearrangement indicator analysis showed H^- hopping events tend to form a large cluster, indicating cooperativity in the H^- diffusion.

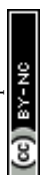


1. INTRODUCTION

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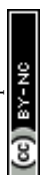
Energy and chemical conversion technologies based on hydrogen transport have become one of the solutions for realizing environmental sustainability.¹ Hydride ions (H^-), which possess a strong reducing ability due to their redox potential of approximately 0.36 V vs. Li^+/Li ,² have been proposed as alternative charge carriers to protons (H^+) for electrochemical applications.^{3,4} Recently, intensive efforts have been devoted to exploring their potential for the development of novel electrochemical devices.^{5–7} Initial studies on H^- conductors primarily focused on alkaline-earth metal hydrides.^{8–10} Significant developments occurred in 2016 with the realization of pure H^- conduction in oxyhydrides, which opened new directions in this field of research. The first example, La–Sr–Li oxyhydride ($\text{La}_{2-x-y}\text{Sr}_{x+y}\text{LiH}_{1+x+y}\text{O}_{3-y}$) with a K_2NiF_4 -type structure, features a framework in which hydride and oxide ions are ordered, thereby forming H^- diffusion pathways within the oxide-based lattice.³ In this compound, tuning the La/Sr ratio allows for control over the H/O ratio and concentration of anion vacancies (V), enabling the modulation of H^- conductivity. The discovery of La–Sr–Li oxyhydrides stimulated the exploration of H^- conductors, particularly mixed-anion compounds such as oxyhydrides.^{11–16} Among the various compounds reported to date, $\text{Ba}_{1.75}\text{LiH}_{2.7}\text{O}_{0.9}$ (BLHO), which adopts the same K_2NiF_4 -type structure as the La–Sr–Li oxyhydride, is of particular interest because of its H^- superionic conductivity that emerges upon undergoing a phase transition at approximately 300 °C.^{17–19}

The ionic conductivity of BLHO was determined by electrochemical impedance spectroscopy (EIS) measurements in a wide temperature range ($200\text{ °C} \leq T \leq 350\text{ °C}$, T is temperature).¹⁷ It exhibited an ionic conductivity of 10^{-5} – 10^{-7} S/cm with an activation energy of approximately 103 kJ/mol at low temperatures ($T \leq 300\text{ °C}$), a drastic increase from 10^{-5} to 10^{-2} S/cm across $300\text{ °C} < T \leq 315\text{ °C}$, and a nearly temperature-independent value at high



temperatures ($315\text{ }^{\circ}\text{C} < T$). This conductivity behavior of BLHO is almost the same as that of Type I superionic conductors. Usually, superionic conductors can be categorized by their phase transition behaviors; Type I involves a discontinuous conductivity jump with a structural (first-order) phase transition, whereas type II shows continuous conductivity change with gradual defect accumulation (higher-order phase transition).^{20,21} The representative AgI shows a conductivity jump by four orders of magnitude at the α - β transition temperature, and the superionic conductivity of the high-temperature α -AgI with the quasi-molten cationic sublattice is characterized by low activation energies (0.1 eV).^{22–24} Therefore, understanding the relationship between the ionic conductivity, crystal structures, and H^- dynamics in BLHO from a solid-state ionic perspective is important.

However, from a crystal structure viewpoint, the phase transition behavior of BLHO differs from that of AgI. BLHO adopts a K_2NiF_4 -type structure in which perovskite and rock-salt layers are alternately stacked along the c -axis. In this structure, the equatorial sites bridging LiX_4 planes, where $X = \text{H}^-$ and V , are either occupied by H^- anions (H_{eq}) or empty (V_{H}). The apical sites are, on the other hand, occupied either by H^- (H_{ap}) or O^{2-} anions (O_{ap}). The occupied and empty Ba sites are denoted as Ba and V_{Ba} , respectively. Those sites ($\text{Ba}/\text{V}_{\text{Ba}}$, the $\text{H}_{\text{eq}}/\text{V}_{\text{H}}$, $\text{H}_{\text{ap}}/\text{O}_{\text{ap}}$, and $\text{Ba}/\text{V}_{\text{Ba}}$) are illustrated in Fig. 1 (a). Under ambient pressure, the stable composition is represented as $(\text{Ba}_{1.75}\text{V}_{0.25})\text{Li}(\text{H}_{1.6}\text{V}_{0.4})_{\text{eq}}(\text{H}_{1.1}\text{O}_{0.9})_{\text{ap}}$, which composition and high vacancy concentration were confirmed by neutron diffraction analysis.¹⁷ While the composition of BLHO includes large amounts of intrinsic H and Ba vacancies, no significant change in composition was observed during the transitions of β - γ - δ phases. As shown in Figs. 1 (a), (b), and (c), there are different anion configurations depending on temperature, defined as β - ($T \leq 300\text{ }^{\circ}\text{C}$), γ - ($300\text{ }^{\circ}\text{C} \leq T \leq 350\text{ }^{\circ}\text{C}$), and δ - ($350\text{ }^{\circ}\text{C} \leq T$) phases from the low-temperature side.¹⁷ The β - γ and γ - δ phase transitions with increasing temperature can be interpreted as the order/disorder transition of $\text{H}_{\text{eq}}/\text{V}_{\text{H}}$ (and also $\text{Ba}/\text{V}_{\text{Ba}}$) and $\text{H}_{\text{ap}}/\text{O}_{\text{ap}}$ sites,



respectively, accompanied by the crystal symmetry change from space groups of $Pnma$ through $Pnma$ to $I4/mmm$. The behavior, where superionic conductivity emerges through changes in the ordering of carrier ions while maintaining the basic framework structure, is akin to that of type II superionic conductors such as β - PbF_2 . They exhibit a gradual transition from an insulating to a conducting state²⁵ with a second-order phase transition.^{20,26,27} Regardless of type I or II, superionic conduction can be characterized in terms of sublattice melting, which involves the disordering of one of the ion sublattices. In the case of BLHO, the equatorial LiX_4 plane can be interpreted as an H^- diffusion layer; thus, the disordered arrangement of $\text{H}_{\text{eq}}/\text{V}_{\text{H}}$ at the equatorial sites implies the occurrence of sublattice melting.

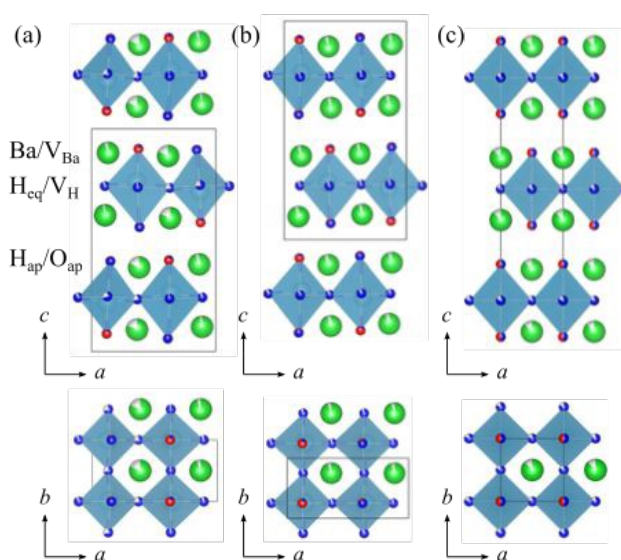


Figure 1. Crystal structures of (a) β -, (b) γ -, and (c) δ -phase BLHO (side and top views). $\text{Ba}/\text{V}_{\text{Ba}}$, Li , $\text{H}_{\text{eq}}/\text{V}_{\text{H}}$, and $\text{H}_{\text{ap}}/\text{O}_{\text{ap}}$ sites are represented by green/white, light blue, blue/white, and blue/red spheres, respectively. Blue octahedrons and black rectangles represent the LiX_6 unit and periodic unit cell, respectively.

In the previous study, an indicative analysis of the H^- dynamics in BLHO was conducted using quasi-elastic neutron scattering (QENS).¹⁷ From the momentum transfer dependence of

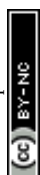


the QENS signal at 315 °C, the self-diffusion coefficient of hydrogen (D_{H}) estimated based on a jump-diffusion model was $3\text{--}4 \times 10^{-7} \text{ cm}^2/\text{s}$. Considering the H^- concentration in BLHO, this value is consistent with the ionic conductivity obtained from EIS, providing evidence that the high conductivity of BLHO arises from fast H^- diffusion. In contrast, the QENS signal at 275 °C, prior to the conductivity jump and β - γ transition, showed a comparable level of H^- diffusivity to that at 315 °C, suggesting that only local H^- motion was accelerated in the β phase, whereas the structural phase transition to the γ phase enabled long-range H^- diffusion that directly contributed to ionic conductivity. These results suggest that the conduction mechanism of H^- changes from the typical jump diffusion model through the β - γ transition; however, they do not detect sublattice melting that characterizes the superionic state.

Therefore, in this study, we performed first-principles molecular dynamics (FPMD) simulations to investigate the relationship between H^- diffusion and BLHO sublattice melting. Since X-ray diffraction experiments provide only information on the average crystal structure, we first constructed several microscopic structural models. The structural models satisfying periodic boundary conditions consists of approximately 200 atoms. Next, we verified the structural models constructed for each phase by comparing their H^- diffusion coefficients and activation energies with those from EIS and QENS data. Then, sub-nanosecond-scale FPMD calculations revealed behavior interpretable as H^- sublattice melting, and rapid H^- diffusion accompanied by collective motion. Comparison with previous studies on typical superionic conductors (PbF_2 , AgI) suggests that the large H^- conductivity of BLHO is likely due to sublattice melting.

2. METHODS

2.1 Models



Before conducting the FPMD simulations, we carefully examined several microscopic structural models of the system. Due to the non-stoichiometric nature of BLHO and the disordered arrangement of ions and vacancies, it is necessary to construct a microscopic supercell model. However, this model cannot be determined uniquely. From among the possible models, an appropriate one can be selected based on the reproducibility of measured quantities such as electronic structure and diffusion properties.

First, considering the pure H^- conducting behavior, the BLHO models should be bandgap semiconductors, which require a charge neutrality condition with respect to Ba^{2+} , Li^+ , H^- , and O^{2-} ions.

As a model that reproduces the experimental composition ratio as faithfully as possible, we selected the β -type $2 \times 4 \times 1$ model shown in Figs. 2 (a) and S1 (a)–(c). In this model, the H atoms and O atoms at the vertex are relatively neatly aligned along the b axis. Using this as a reference, we created three variants by adding small amount of Ba, H, and O atoms, that is, based on this model: $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$, $\text{Ba}_{57}\text{Li}_{32}\text{H}_{90}\text{O}_{28}$, and $\text{Ba}_{58}\text{Li}_{32}\text{H}_{90}\text{O}_{29}$. The composition ratios and numbers of H_{eq} and Ba vacancies of the three models are listed in Table S1 of the Supplementary Information (SI). The configurations and cell-optimized structures are shown in Figs. S2 (a), (b), and (c). In all models, the Li–O and counter side Li– H_{ap} distances varied between 1.85–2.43 and 1.93–3.41 Å, respectively, resulting in the formation of LiH_3O tetrahedrons and free H_{ap} atoms.



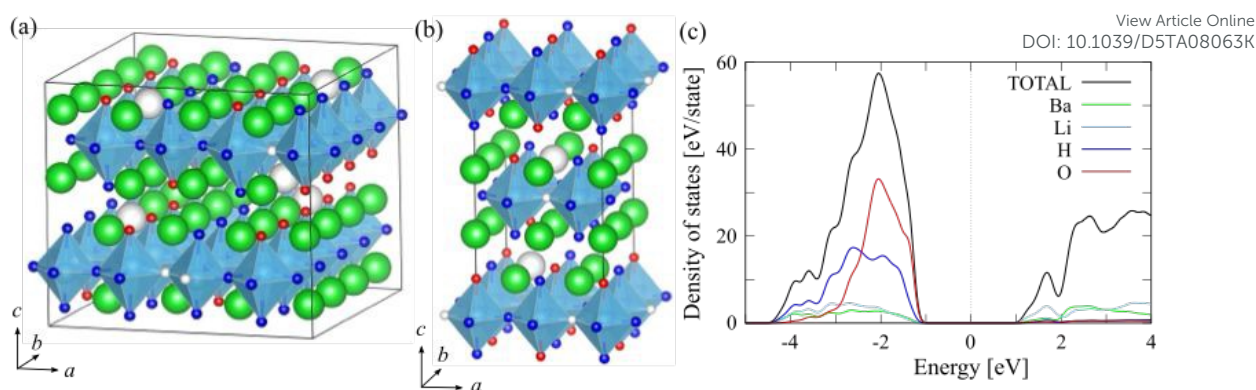


Figure 2. Crystal structures of (a) β -like $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$ and (b) δ -like $\text{Ba}_{14}\text{Li}_8\text{H}_{22}\text{O}_7$. Ba, Li, H, and O atoms are depicted as green, light blue, blue, and red spheres, respectively. V_H and V_Ba are depicted as small and big white spheres, respectively. (c) PDOSs of the δ -like $\text{Ba}_{14}\text{Li}_8\text{H}_{22}\text{O}_7$ model. The black, green, light blue, blue, and red lines represent the total, Ba, Li, H, and O contributions, respectively. The dotted line indicates the Fermi level located at the middle of bandgap.

Next, we move on to constructing the δ -like model. Figures 2 (b) and S3 (a)–(c) show the reference model of $\text{Ba}_{14}\text{Li}_8\text{H}_{22}\text{O}_7$ ($\text{Ba}_{1.75}\text{LiH}_{2.75}\text{O}_{0.875}$). In this model, the H and O atoms at the apical sites are alternately aligned along both the a - and b -axes of the unit cell. Based on this δ -like $2 \times 2 \times 1$ model, we constructed three composition-adjusted models in a $4 \times 4 \times 1$ supercell: $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$, $\text{Ba}_{57}\text{Li}_{32}\text{H}_{90}\text{O}_{28}$, and $\text{Ba}_{58}\text{Li}_{32}\text{H}_{90}\text{O}_{29}$, whose configurational and cell-optimized structures are shown in Figs. S4 (a), (b), and (c). Similar to the β -like models, the Li–O and counter side Li– H_ap distances varied between 1.81–1.93 and 2.73–3.69 Å, respectively, which results in the formation of LiH_3O tetrahedrons. The large variation of distances might be related to the observed large B factors of Li atoms in BLHO, $B_\text{Li} = 1.90$ (RT), 6.1 (314 °C), and 6.4 (368 °C) Å², as shown in ref. 17. As expected from the experimental phases, the optimized energies of the δ -like models are higher than those of the β -like models by approximately 0.12–0.16 eV/Li. In this study, while the local structures and vacancy



concentration of constructed models do not contradict with high-precision neutron diffraction analysis,¹⁷ it is not possible to verify their thermodynamic characteristics such as vacancy formation energy. Therefore, verification from this perspective remains a topic for future research.

The optimized cell parameters and calculated bandgap energies are listed in Table S2 in SI. Slight compositional changes do not significantly affect the calculated values. Although the cell parameters are slightly overestimated compared with those of the β -phase, which is typical for standard density functional theory (DFT) calculations. All models show electronic bandgap energies of 2.2–2.7 eV. Considering that DFT calculations with semilocal exchange-correlation functional tend to underestimate the bandgap value, the calculated values suggest the semiconducting or insulating electronic conductivity of BLHO. Additionally, we confirmed with spin-polarized calculations that the total magnetic moments of all the relaxed structures are negligibly small.

Figure 2 (c) shows the projected densities of states (PDOSs) of the δ -like $\text{Ba}_{14}\text{Li}_8\text{H}_{22}\text{O}_7$ model, which shows typical property in BLHO structure models. The H-1s and O-2p orbitals mainly contribute to the valence band, and the Li-2s orbitals contributes to the conduction band. Regardless of the bonding and antibonding states, all orbitals with significant H-1s component are occupied, which can be interpreted as an H^- like character. This valence band property is almost the same as that of K_2NiF_4 -type La–Sr–Li and Ba–Sc oxyhydride.^{28–31}

To discuss the effective charge distributions in BLHO, we assigned an electronic charge density to each atom using the Bader charge analysis.³² Positive and negative ions are easily distinguished from the Bader charges, as listed in Table 1. These values are close to the formal ionic charges (Ba^{2+} , Li^+ , H^- , and O^{2-}), and the fractional parts of the charge value represent a small amount of covalent properties. In addition, the Bader charges of Li, H, and O in BLHO



are similar to those in La–Sr–Li oxyhydrides,^{28,29} indicating that the ionic and bonding characteristics of K₂NiF₄-type oxyhydrides are common.

Table 1. Average values of Bader charge in the δ -like Ba₁₄Li₈H₂₂O₇ model.

Atom	Charges [e]
Ba	+1.37
Li	+0.85
H	−0.73
O	−1.45

Based on the above preparative calculations, we adopted the simplest models, β - and δ -like Ba₅₆Li₃₂H₈₈O₂₈, as the initial structures used in the FPMD simulations. Namely, we conducted long term FPMD simulations using two possible configurations mimicked β and δ crystal structures. Because the crystallographic difference between the β and γ phases can be attributed to the partial occupancy ratios of the equatorial anion sites, we consider that the phase transition from β to γ can be included automatically in the FPMD simulations at high temperatures, even though the initial structures are started from β -like models. In addition, there are numerous possible configurations for three partial occupancy sites: H_{eq}/V_H, Ba/V_{Ba} and H_{ap}/O_{ap}. For the partial occupancy effect at H_{eq}/V_H site, the H[−] hopping between H_{eq}/V_H sites frequently occurred at higher temperatures and suitable statistical ensemble can be accounted for to some context. While for the Ba/V_{Ba} and H_{ap}/O_{ap} sites, this is not the case owing to the limited duration of the FPMD simulations, necessitating further investigation in future studies.

2.2 Computational details



The β - and δ -like models as shown in Figs. 2 (a) and (b), respectively, were constructed from the refined structural parameters of β - and δ -BLHO.¹⁷ The VESTA package was used to visualize the crystal structures and atom coordinates in this study.³³

To obtain optimized atomic configurations and cell parameters, band-gap energies, and PDOSs, the Quantum ESPRESSO (QE) code,^{34,35} a plane-wave basis within the ultrasoft pseudopotential framework,^{36,37} was used to perform spin-unpolarized DFT calculations. It was confirmed that spin-polarized calculations, performed by starting from various initial spin configurations, yielded the same non-spin states that obtained in spin-unpolarized calculations. This indicates that the H^- property is maintained throughout the nudged elastic band (NEB) calculation. The electronic configurations were $1s^1$ for H, $1s^2 2s^1$ for Li with nonlinear core correction (NLCC),³⁸ $2s^2 2p^4$ for O with NLCC, and $5s^2 5p^6 6s^2$ for Ba with NLCC; the pseudopotentials of H.pbe-rrkjus_psl.1.0.0.UPF, Li.pbe-sl-rrkjus_psl.1.0.0.UPF, O.pbe-n-rrkjus_psl.1.0.0.UPF, and Ba.pbe-spn-rrkjus_psl.1.0.0.UPF from the QE pseudopotential database were used.^{39,40} All DFT calculations were performed using the Perdew–Burke–Ernzerhof exchange–correlation functional.⁴¹ The cutoff energies were set to 40/320 Ry (1 Ry = 13.606 eV) for the wave functions/augmented charge. Only for the cell optimization procedure, they increased to 60/480 Ry to ensure convergence. Converged k-point samplings of $4 \times 4 \times 2$ and $2 \times 2 \times 2$ were adopted for the $2 \times 2 \times 1$ and $4 \times 4 \times 1$ cells, respectively. The atomic positions and cell parameters were relaxed until the residual forces and stresses were less than 10^{-3} Ry/bohr (1 bohr = 0.52918 Å) and 0.5 kbar (1 bar = 100 Pa), respectively. For the Bader charge analysis, we used a program distributed by the Henkelman group.^{42,43}

The NEB method,^{44,45} which can determine the minimum energy path between two stable points, was used to evaluate the activation energies of H^- migration. The NEB method was applied to a discretized path using eight images. The start- and end-point configurations were constructed by replacing the H atom at the next site. All atoms in the start and end images were



relaxed in advance of NEB calculations, whereas those in the intermediate images were relaxed until their forces converged within 0.05 eV/Å. The electronic structure calculation conditions were the same as those used in the QE calculations. The cell parameters were fixed at the experimental values. All NEB calculations were performed using a representative δ -like model of Ba₅₆Li₃₂H₈₈O₂₈.

For FPMD simulations, we used the Vienna Ab initio Simulation Package (VASP),^{46–48} a plane-wave basis within the projector-augmented-wave method^{49,50} with a cutoff energy of 400 eV for the wave functions. A single Γ -centered k-point and time step of 2 fs were adopted in spin-unpolarized DFT calculations. In hydrogen containing materials, shorter timestep (1 fs) is typically chosen to describe rapid H vibrations, such as the OH stretching mode (frequency of 3000 cm^{–1}). In hydride ion conductors, however, longer timescale of 2 fs can be adopted²⁹ because the maximum phonon frequency is significantly less than the OH frequencies (e.g., approximately 1600 cm^{–1} in La–Li oxyhydride²⁸). The relaxed β - and δ -like structures were assigned an initial temperature of 300 K according to the Boltzmann distribution and then heated to the desired temperatures in the NVT ensemble with a Nose–Hoover thermostat.^{51,52} FPMD simulations were conducted for approximately 200 ps under thermal equilibrium conditions at $T = 500, 700, 900,$ and 1100 K. The self-diffusion coefficients, Lindemann indices, and van Hove time correlation functions were then calculated from the atomic trajectories obtained from the FPMD simulations.

The self-diffusion coefficient of H atoms was calculated by the mean square displacement (MSD) over interval time Δt ,^{29,53}

$$D = \lim_{\Delta t \rightarrow \infty} \frac{1}{2Nd\Delta t} \sum_{i \in H}^N \langle |\mathbf{r}_i(t+\Delta t) - \mathbf{r}_i(t)|^2 \rangle, \quad (1)$$

where d , N , $\mathbf{r}_i(t)$, and bracket $\langle \cdots \rangle$ represent the dimension of diffusion system, total number of diffusion ions, position vector of i -th atom at time t , and averaging over time t , respectively.



The ab -plane and c -axis components, D_{ab} and D_c , respectively, were separated, reflecting the layer-like structure of BLHO.

The Lindemann index δ_L ,^{54–56} which measures the atomic vibrational amplitude, was calculated as follows:

$$\delta_L(M) = \frac{1}{N(N-1)} \sum_{\substack{i \neq j \\ i, j \in M}}^N \frac{\sqrt{\langle r_{ij}^2(t) \rangle - \langle r_{ij}(t) \rangle^2}}{\langle r_{ij}(t) \rangle} \quad (M = \text{H, O, Li, or Ba}), \quad (2)$$

$$r_{ij}(t) = |\mathbf{r}_i(t) - \mathbf{r}_j(t)|, \quad (3)$$

where M and $r_{ij}(t)$ represent the atomic species and the distance between the i - and j -th atoms, respectively. If δ_L reaches a certain value, it is presumed that fluctuations cannot increase without damaging or destroying the lattice structure. Thus, δ_L is commonly used as an indicator of solid–liquid transition in the field of physical chemistry. The critical threshold, empirically $0.10 \lesssim \delta_L \lesssim 0.15$, has been found to be relatively independent of the types of materials and computational models.⁵⁶ Although it has not been established that δ_L can be used as an indicator of the entropy change behavior observed in superionic conductors, we suggest the usefulness of this index to discuss sublattice melting in BLHO.

The distinct part of van Hove correlation function G_d ,^{57–60} which describes the time-space correlation between selected atoms, was calculated as functions of radial coordinate r and time distance Δt .⁵³

$$G_d(r, \Delta t) = \frac{1}{4\pi r^2 N \rho} \left\langle \sum_{\substack{i \neq j \\ i, j \in \text{H}}}^N \delta(r - |\mathbf{r}_i(t+\Delta t) - \mathbf{r}_j(t)|) \right\rangle_t, \quad (4)$$

where ρ and δ are the average density and Dirac delta function, respectively. In the short-time limit, the above equation becomes the radial distribution function: $G_d(r, \Delta t = 0) = g(r)$, whereas in the long-time limit, it becomes independent of the radial coordinates: $G_d(r, \Delta t \rightarrow \infty) = 1$.



We analyzed the number of hopping events and their cluster sizes using R_i ,⁶¹ (rearrangement index), an indicator of net atomic displacement. This index is characterized by a typical reconfiguration time interval, Δt :

$$R_i(t, \Delta t) = \sqrt{\langle (\mathbf{r}_i(t) - \langle \mathbf{r}_i(t) \rangle_{t-\Delta t})^2 \rangle_{t+\Delta t} \langle (\mathbf{r}_i(t) - \langle \mathbf{r}_i(t) \rangle_{t+\Delta t})^2 \rangle_{t-\Delta t}}, \quad (5)$$

where brackets $\langle \dots \rangle_{t \pm \Delta t}$ represent averaging over time t to $t \pm \Delta t$. When the square root of this quantity exceeds a typical interatomic distance $\sqrt{R_i} \geq d_1$, we consider that a hopping event has occurred. We selected the value of Δt as 0.3 ps, which is the typical time for H atom hopping; the dependence on the Δt value is discussed later. Collective diffusion of hydride ions is characterized by strings and clusters. A string is defined as a sequence of consecutive jump events of a single atom, bearing resemblance to the chain-like structures observed in molecular dynamics trajectory animations. In order to capture the collective motion of multiple ions occurring nearly simultaneously, a cluster is defined as strings that approach each other within a threshold distance d_1 . The term “cluster size” is used to denote the number of atoms participating in the collective motion. The cluster size is defined as the number of atoms participating in the cluster. Analyses based on R_i have been reported in studies of slow diffusion, such as cage jumps associated with glass transitions⁶² and supercooled liquids.⁶³

3. RESULTS

Figure S5 shows the calculated MSDs of H obtained from temperature-controlled FPMD simulations. At the lowest temperature of 500 K, sufficient H-hopping events are not obtained during the simulation, and the values of D at 500 K contain errors. At higher temperatures, the diffusion process in the ab plane corresponds to H hopping among the equatorial sites, which is consistent with the experimental results of the crystal structure analysis. Moreover, at 1100 K, the δ -like model showed that diffusion in the c -direction is comparable to that in the ab -



plane, as shown in Fig. S5 (h), although BLHO is expected to melt in this high-temperature region.

Figure 3 shows the diffusion coefficients D_{ab} and D_c obtained from the slopes of the MSD data. It should be noted that the simulation length did not reach the diffusion regime where MSD is clearly linear with respect to time, and consequently, the diffusion barrier could not be accurately estimated. However, it is possible to compare qualitatively the differences arising from the phase variations. In the high-temperature region ($700\text{ K} \leq T \leq 1100\text{ K}$), the D_{ab} values of the β model are almost the same as those of the δ model, indicating no significant difference between the β and δ phases regarding ab -plane diffusion and conduction. The D_c values are distinctly lower for the β -like model than for the δ -like model at temperatures above 900 K. This suggests that the alternating arrangement of the apical H and O atoms hinders c -axis diffusion. Furthermore, the interpolation of D_{ab} of the δ model reproduces the reported value ($D = 3\text{--}4 \times 10^{-7}\text{ cm}^2/\text{s}$ at $315\text{ }^\circ\text{C}$) by QENS analysis,¹⁷ and that of the β model is also consistent with the previous value. This agreement strongly suggests that the FPMD simulation of the constructed microscopic models is reliable for discussing the sublattice melting as well as dynamical aspects of H^- diffusion in BLHO.

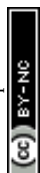
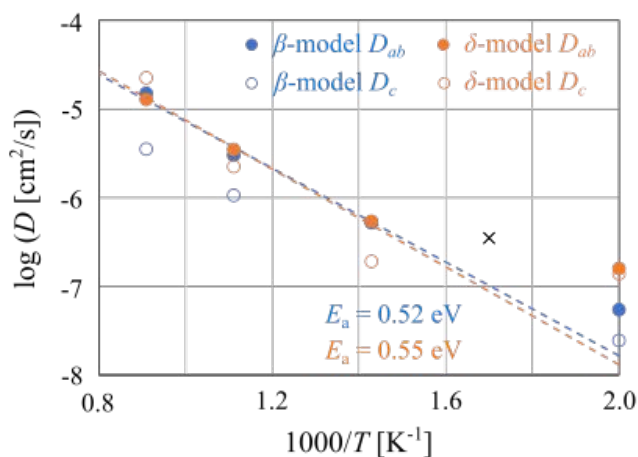


Figure 3. Diffusion coefficients D of H as a function of inverse temperature. Blue and orange represent β - and δ -like models, respectively. Solid and open circles represent the D of the ab -plane and c -axis components, respectively. Dashed lines represent the corresponding activation energies of the ab -components in the high-temperature region ($700\text{ K} \leq T \leq 1100\text{ K}$). Cross mark represents the D value obtained by QENS analysis.¹⁷

The estimated activation energies E_a from D_{ab} in the high-temperature region ($700\text{ K} \leq T \leq 1100\text{ K}$) are 0.52 and 0.55 eV for the β - and δ -models, respectively. These values differ from those obtained from experimental ionic conductivity: $E_a = 1.1\text{ eV}$ (β phase) and 0.0 eV (γ phase). The comparison indicates that the rate-determining process of macroscopic conductivity is not the ab -plane diffusion of H^- but rather is related to the ionic diffusion of H^- and O^{2-} along the c -direction, their diffusion at crystallite interfaces, or structural transition. In low temperature region, the c -direction diffusion is one of the possible rate-determining steps. The reliable large-scale simulations should be conducted to obtain the activation energy of c -direction diffusion quantitatively in the future.

The activation energies of H^- diffusion from the FPMD simulations are associated with elementary processes by NEB calculations. We should note that, in a disordered system, it is necessary to perform numerous NEB calculations for various configurations to obtain the average activation energy. Hence, the NEB results can be regarded as the rough estimation of the activation energies and supporting data for FPMD calculations. Figure 4 (a) and (b) show the energy profiles along the ab -plane and c -axis H migration pathways, respectively, assuming V_H -hopping processes within the δ -like model. We obtained several activation energies depending on the local bonding environment for ab -plane H migration or H hopping between the equatorial sites. For example, in the H1 migration path, the initial Li1–H1 bond length is large (LiX_4 like) and rather unstable, which resulted in a slightly higher energy at the start point



than that at the end. The H2 migration path, hopping between LiX_5 , showed a high activation energy of approximately 0.4 eV. Finally, the energy range between the minimum (end point of the H1 path) and maximum (transition state of the H2 path) NEB images was 0.50 eV, which was consistent with the activation energy of 0.55 eV obtained from D_{ab} of the δ -model.

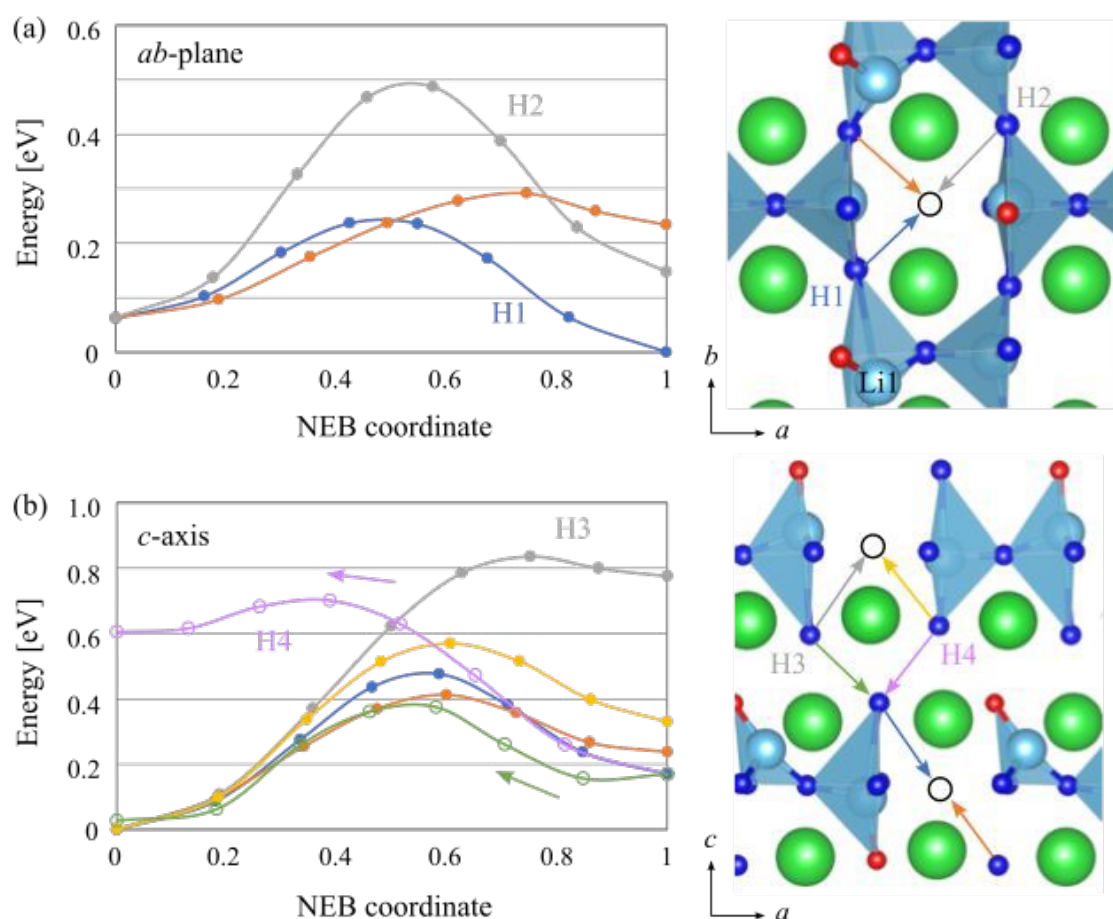


Figure 4. Energy profiles along (a) *ab*-plane and (b) *c*-axis for H migrations. Solid and open circles represent the relative energies from the NEB calculations, and solid lines are guides for the eye, the colors of which correspond to the migration paths indicated by the colored arrows in the subset structures. In the purple and green migration paths, the forward H moves to the next vacancy before the backward H starts moving.



For *c*-axis diffusion or H hopping between the apical and equatorial sites, we obtained several activation energies that depended on the local bonding environment. In general, the relative energy of the V_H at the apical site was approximately 0.2–0.3 eV higher than that at the equatorial site. Occasionally, a high-energy configuration occurred, such as the end points of the H3 and H4 paths, which indicated that Li polyhedrons constructed only from H^- , that is, LiH_4 and LiH_6 , were quite unstable. Finally, the energy range between the minimum and maximum energy configurations was 0.82 eV, which was consistent with the activation energy of 0.78 eV obtained from D_c of the δ -model in the high-temperature region ($700\text{ K} \leq T \leq 1100\text{ K}$).

4. DISCUSSION

In this section, we discuss whether sublattice melting, the disordering of one of the ion sublattices, occurs in BLHO. This phenomenon is usually characterized by a peak in the specific heat;²⁰ however, the available FPMD data for temperatures are rather sparse. Alternatively, we use the Lindemann index δ_L as an indicator of liquid-like behavior. Furthermore, we discuss the behavior of two-body correlations and multi-ion concerted diffusion of H^- in terms of the van Hove correlation function and hopping cluster analysis.

Figure 5 shows the δ_L of each atom in BLHO from the FPMD simulations. At the low temperature of 500 K, there is no significant difference between the β and δ models, and all values are lower than the critical threshold of the solid–liquid transition. At the middle temperature of 700 K, $\delta_L(H)$ exceeded the threshold for both models, suggesting sublattice melting at the anion equatorial sites of H_{eq}/V_H . It should be noted that only the equatorial site contribution, $\delta_L(H_{eq})$, is larger than the plotted values because the calculated $\delta_L(H)$ also includes the fluctuations of the apical sites (H_{ap}). Because the critical temperature exceeding the threshold of the solid–liquid transition is consistent with the β – γ transition temperature (T

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$\approx 300\text{ }^{\circ}\text{C}$), we consider the results of $\delta_L(\text{H})$ as theoretical suggestions for sublattice melting. View Article Online
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BLHO is one of the superionic conductors.

At the high temperature of 900 K, $\delta_L(\text{H})$ significantly increased, and that of the δ model exceeded 0.2. The $\delta_L(\text{H})$ difference between the β and δ models can be attributed to c -axis diffusion. The $\delta_L(\text{Li})$ values are also comparable to the solid–liquid threshold. At the highest temperature of 1100 K, all the δ_L values of the δ model exceeded the threshold, indicating that BLHO cannot maintain the solid-state crystal structure, that is, a typical solid–liquid transition occurs below this temperature. Finally, it should be noted that the indication of sublattice melting is qualitative. Toward the quantitative analysis, it is desirable to obtain the dynamical property in the occurrence of sublattice melting, which discussed in the last paragraphs.

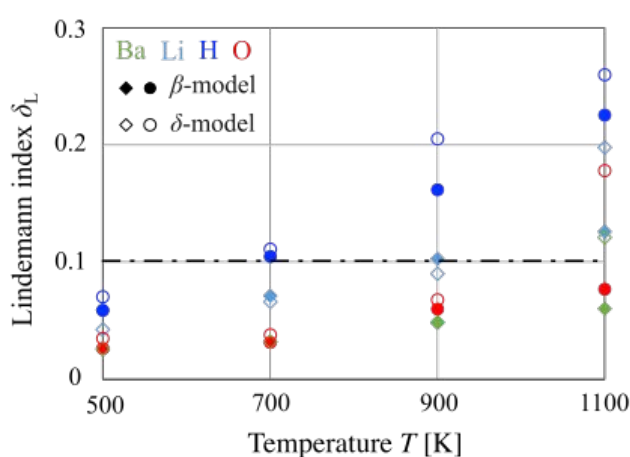


Figure 5. Lindemann indices δ_L as a function of temperature T . Solid and open symbols represent the β - and δ -like models, respectively. Green, light blue, blue, and red colors represent the Ba, Li, H, and O atoms, respectively. The critical threshold of the solid–liquid transition ($= 0.1$) is indicated by the black dash-dotted line.

Several studies suggest that sublattice melting affects ion diffusion through significant coupling with anharmonic phonons and parent lattice vibrations.^{64,65} Besides the thermodynamic analysis using the Lindemann index, we briefly discuss the dynamic properties



related to the sublattice melting of H^- from here. Figure 6 (a) shows the radial distribution function $g(r)$ of the δ model at the four controlled temperatures. At the lowest temperature of 500 K, the first peak position of 2.86 Å can be assigned to the first nearest-neighbor distance between equatorial sites, $d_1(\text{H}_{\text{eq}}-\text{H}_{\text{eq}}) \approx 2.8$ Å, and that between equatorial and apical sites, $d_1(\text{H}_{\text{eq}}-\text{H}_{\text{ap}}) \approx 3.0$ Å. The shoulder peak at approximately 4.0 Å corresponds to the second nearest-neighbor distance between equatorial sites, $d_2(\text{H}_{\text{eq}}-\text{H}_{\text{eq}}) \approx 4.0$ Å, and the first nearest-neighbor distance between apical sites, $d_1(\text{H}_{\text{ap}}-\text{H}_{\text{ap}}) \approx 3.8$ Å. The second peak at 5.02 Å corresponds to the second nearest-neighbor distance between the equatorial and apical sites, $d_2(\text{H}_{\text{eq}}-\text{H}_{\text{ap}}) \approx 5.0$ Å. At the increased temperatures of 700–1100 K, the separation between the first and shoulder peaks becomes ambiguous. This continuous distribution of H^- is consistent with the sublattice melting obtained from the Lindemann index.

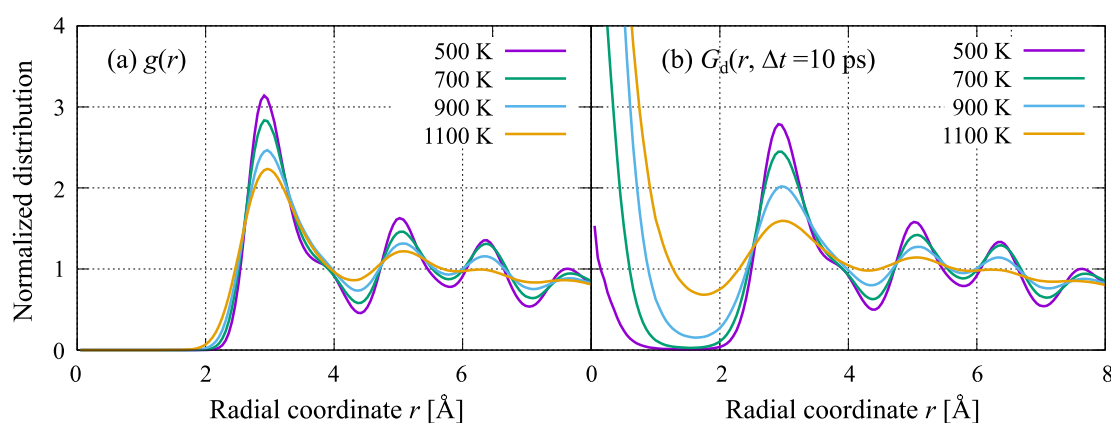


Figure 6. (a) Radial distribution function $g(r)$ and (b) van Hove correlation function $G_d(r, \Delta t = 10 \text{ ps})$ of the δ -like model. Magenta, green, blue, and orange lines represent those obtained at $T = 500, 700, 900$, and 1100 K, respectively.

Figure 6 (b) shows the van Hove time-correlation function G_d set the interval time Δt to 10 ps (The systematic G_d plots of the β and δ models with gradually changing interval times are shown in Fig. S6 (a–h)). At all temperatures, the correlation holes at $r = 0$ quickly annihilated



and peak growths continued for the time interval of several tens of picoseconds, which can be interpreted as the V_H created by a certain H-hopping event being immediately occupied by another H-hopping event. This is strong evidence of the concerted-like hopping migration, as discussed in a previous study.⁵³ The analysis reveals that most H^- migrate in a highly concerted fashion and multiple ions hop simultaneously into their nearest sites within a few picoseconds. Finally, at all temperatures, the peak growths continued even at the maximum time of the FPMD simulations.

H atoms exhibit a string-like MD trajectory, which was analyzed using the rearrangement indicator following Mohn et al.⁶¹ In this study, we selected the reconfiguration time interval $\Delta t = 0.3$ ps, which was estimated from the MD trajectory as the timescale for site-to-site hopping of H^- ion. It was confirmed that the resulting behavior is not sensitive to the choice of Δt (see Fig. S7), which suggests that the collective motion can be reliably captured by our analysis. All the strings appeared in the entire MD trajectory at $T = 700$ K are shown in Fig. 7 (a), as typically found in supercooled liquids⁶² and glasses.⁶³ Number of strings constituting a cluster, or the cluster size, is now analyzed. Figures 7 (b) and S7 (a–d) show distribution of the number of hopping events that occur forming a cluster of size n . As the temperature increases, hopping tend to occur forming a larger cluster, which is significant above the sublattice melting temperature ($T \geq 700$ K). This indicates increased cooperativity in the H diffusion. In addition, the interlayer H migrations tend to occur through the Ba vacancy, V_{Ba} , indicating important role played by the V_{Ba} configuration. Note that one configuration as appeared in the supercell model was examined in this study, but sampling the configuration would be an important target of future study. Next, we analyzed correlation between H-movement in the c -axis direction and magnitude of Li-displacement. The correlation is appreciable although not so strong, as shown in Figs. S8 (a–c), which indicates contribution of Li atoms in the collective H diffusion. Analysis of the MD trajectory have thus revealed that diffusion in BLHO is characterized by



the correlated motion of H atoms accompanying the displacement of Li atoms. We consider that a large number of V_H (and V_{Ba}) is essential factors to induce the dynamical ion-lattice correlation as well as sublattice melting. It should be noted that the reason why BLHO can introduce such amounts of vacancies is not known in terms of solid-state chemistry; therefore, the thermodynamic stability of BLHO and defect properties should be clarified in the future.

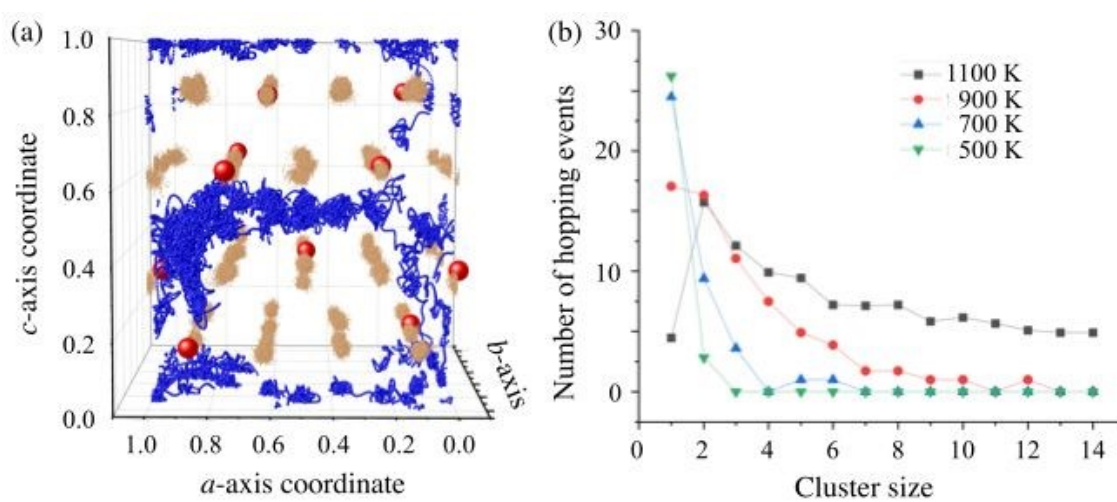


Figure 7. (a) Visualization of H⁻ diffusion at $T = 700$ K. Blue strings represent the H atom trajectories with cluster sizes ≥ 3 . Orange strings and red spheres represent the Ba atom trajectories and Ba vacancies, respectively. (b) Number of hopping events plotted against the cluster size involved in the hop. Green, blue, red, and black symbols represent the values obtained at $T = 500$, 700 , 900 , and 1100 K, respectively.

5. CONCLUSIONS

We investigated the H⁻ conduction mechanism and anion sublattice melting in BLHO using first-principles electronic structure calculations and molecular dynamics simulations. First, we examined several microscopic structure models, such as $Ba_{56}Li_{32}H_{88}O_{28}$, $Ba_{57}Li_{32}H_{90}O_{28}$, and $Ba_{58}Li_{32}H_{90}O_{29}$, where the H and O atoms at the apical sites were ordered by β - and δ -phase like structures. The constructed models showed bandgaps of 2.2–2.7 eV, occupied H-1s orbitals,



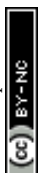
and Bader charges close to their formal ionic charges (Ba^{2+} , Li^+ , H^- , and O^{2-}), consistent with the pure H^- conduction and electrochemical H^- carriers observed in experiments. Second, using the constructed β - and δ -like models of $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$, FPMD simulations were performed for several hundred picoseconds under temperature-controlled conditions at $T = 500, 700, 900$, and 1100 K. The obtained D_{ab} of H was in good agreement with that obtained from the QENS measurement, indicating that the local-scale H^- dynamics were well reproduced. The activation energies of D_{ab} (0.52–0.55 eV in both models) and D_c (0.78 eV in the δ model) were associated with individual H^- migration paths, with energy barriers of 0.50 and 0.82 eV for H^- hopping between equatorial sites and between apical and equatorial sites, respectively. Furthermore, the Lindemann index of H exceeded the threshold for a solid–liquid transition at $T \geq 700$ K, suggesting the sublattice melting of H^- equatorial sites. Above the sublattice melting temperature, the van Hove correlation function and rearrangement indicator analysis showed the peak growth at correlation hole position increased for several tens of picoseconds and H^- hopping events tend to form a large cluster, respectively, indicating cooperativity in the H^- diffusion.

ASSOCIATED CONTENT

Author contributions

Jun Haruyama: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, software, validation, visualization, writing – original draft.

Fumitaka Takeiri: conceptualization, writing – review & editing. Genki Kobayashi: conceptualization, supervision, writing – review & editing. Osamu Sugino: conceptualization, formal analysis, funding acquisition, methodology, resources, software, supervision, validation, visualization, writing – review & editing.



Conflicts of interest

There are no conflicts to declare.

Data availability

Data for this article, including VASP and QE inputs and trajectory analysis programs are available at the ISSP data repository at https://isspns-gitlab.issp.u-tokyo.ac.jp/j-haruyama/BLHO_DFT-MD. The data supporting this article have been included as part of the SI. Supplementary information: Crystal structures of the β -like $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$ model. Table of the number of atoms, composition ratios, and numbers of H_{eq} and Ba vacancies in the composition change models. Optimized structures of β -like $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$, $\text{Ba}_{57}\text{Li}_{32}\text{H}_{90}\text{O}_{28}$, and $\text{Ba}_{58}\text{Li}_{32}\text{H}_{90}\text{O}_{29}$ models. Crystal structure of the δ -like $\text{Ba}_{14}\text{Li}_8\text{H}_{22}\text{O}_7$ model. Optimized structures of δ -like $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$, $\text{Ba}_{57}\text{Li}_{32}\text{H}_{90}\text{O}_{28}$, and $\text{Ba}_{58}\text{Li}_{32}\text{H}_{90}\text{O}_{29}$ models. Table of the mimicked phases, number of atoms, optimized lattice constants, and calculated energy gaps of the composition change models. Mean square displacements of H obtained by the β - and δ -like models at controlled temperatures. van Hove correlation function as a function of the radial coordinate obtained by the β - and δ -like models at controlled temperatures. The number of hopping events as functions of cluster size and rearrangement time interval. Correlation between Li and H motions.

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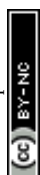
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Data availability

Data for this article, including VASP and QE inputs and trajectory analysis programs are available at the ISSP data repository at https://isspns-gitlab.issp.u-tokyo.ac.jp/j-haruyama/BLHO_DFT-MD. The data supporting this article have been included as part of the SI. Supplementary information: Crystal structures of the β -like $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$ model. Table of the number of atoms, composition ratios, and numbers of H_{eq} and Ba vacancies in the composition change models. Optimized structures of β -like $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$, $\text{Ba}_{57}\text{Li}_{32}\text{H}_{90}\text{O}_{28}$, and $\text{Ba}_{58}\text{Li}_{32}\text{H}_{90}\text{O}_{29}$ models. Crystal structure of the δ -like $\text{Ba}_{14}\text{Li}_8\text{H}_{22}\text{O}_7$ model. Optimized structures of δ -like $\text{Ba}_{56}\text{Li}_{32}\text{H}_{88}\text{O}_{28}$, $\text{Ba}_{57}\text{Li}_{32}\text{H}_{90}\text{O}_{28}$, and $\text{Ba}_{58}\text{Li}_{32}\text{H}_{90}\text{O}_{29}$ models. Table of the mimicked phases, number of atoms, optimized lattice constants, and calculated energy gaps of the composition change models. Mean square displacements of H obtained by the β - and δ -like models at controlled temperatures. van Hove correlation function as a function of the radial coordinate obtained by the β - and δ -like models at controlled temperatures. The number of hopping events as functions of cluster size and rearrangement time interval. Correlation between Li and H motions.

